RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Chapter 9A: Integration Techniques

SYLLABUS INCLUDES

- Integration of $\mathrm{f}'(x)[\mathrm{f}(x)]^n$, $\mathrm{f}'(x)\mathrm{e}^{\mathrm{f}(x)}$, $\sin^2 x$, $\cos^2 x$, $\tan^2 x$, $\dfrac{1}{a^2+x^2}$, $\dfrac{1}{\sqrt{a^2-x^2}}$, $\dfrac{1}{a^2-x^2}$ and $\dfrac{1}{x^2-a^2}$
- Integration by a given substitution
- Integration by parts

PRE-REQUISITES

- Basic Trigonometry
- Partial Fractions
- Differentiation Techniques

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- 3.3 Integration by Substitution or Change of Variable
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1 INTRODUCTION

We have seen that differentiating a function with respect to its variable gives us its derivative, i.e. the function that tells us the instantaneous rate of change of the original function with respect to its variable. This raises the question of whether it is possible to recover the original function, given its derivative. Just as differentiation is a process that gives you the derivative of a function, integration as introduced here will be a process that seeks to give you the original function given its derivative, albeit with some additional information required. Here, integration is viewed as **anti-differentiation**, and the functions obtained after integration are known as **anti-derivatives**.

In the next chapter, we will then see how the processes of differentiation and integration are linked via the fundamental ideas of limits and use these principles to show how, just as differentiation may be used to find gradients of curves, integration may also be used to find areas and volumes.

1.1 Integration as the Reverse Process of Differentiation

One basic way to perform integration is to modify the integrand (the expression that is to be integrated), such that it represents the derivative of a common function.

The ability to integrate a function in such a manner depends primarily on recognizing it as the derivative of another function.

Exercise: Given the following expression for $\mathrm{f}(x)$, find an expression $\mathrm{F}(x)$ such that $\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{F}(x) = \mathrm{f}(x)$.

$\mathrm{f}(x)$	$\mathrm{F}(x)$	$\mathrm{f}(x)$	$\mathrm{F}(x)$
1	x	e^x	e^x
$2x$	x^2	$\cos x$	$\sin x$
$3x^2$	x^3	$\sec^2 x$	$\tan x$
x^3	$\frac{x^4}{4}$	$\frac{1}{x}$	$\ln x$

1.2 Indefinite Integral

1.2.1 Notation

Suppose f and F are two functions related as follows: $\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{F}(x)) = \mathrm{f}(x)$.

Then, for any constant c, $\frac{\mathrm{d}}{\mathrm{d}x}\left[\mathrm{F}(x) + c\right] = \mathrm{f}(x)$.

Hence, we have $\mathrm{F}(x) + c = \int \mathrm{f}(x)\mathrm{d}x$.

We call
- $\int \mathrm{f}(x)\ \mathrm{d}x$ the indefinite integral of f with respect to x,
- c the arbitrary constant of integration,
- $\mathrm{f}(x)$ the integrand,
- $\int$ the integral sign.

For example, we have

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}(x-3)^2 + 3\right) = x - 3 \text{ and } \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}(x-3)^2 + 300\right) = x - 3,$$

therefore $\int (x-3)\ \mathrm{d}x = \frac{1}{2}(x-3)^2 + c$.

1.2.2 Basic Rules of Integration

(i) $\int \left[\mathrm{f}(x) \pm \mathrm{g}(x)\right]\mathrm{d}x = \int \mathrm{f}(x)\ \mathrm{d}x \pm \int \mathrm{g}(x)\ \mathrm{d}x$

(ii) $\int k\mathrm{f}(x)\ \mathrm{d}x = k\int \mathrm{f}(x)\ \mathrm{d}x$, where k is a constant, $k \neq 0$

(iii) $\int \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{f}(x)\right)\ \mathrm{d}x = \mathrm{f}(x) + c$

(iv) $\frac{\mathrm{d}}{\mathrm{d}x}\left(\int \mathrm{f}(x)\ \mathrm{d}x\right) = \mathrm{f}(x)$

Note : $\int \left[\mathrm{f}(x)\mathrm{g}(x)\right]\mathrm{d}x \neq \int \mathrm{f}(x)\ \mathrm{d}x \bullet \int \mathrm{g}(x)\ \mathrm{d}x$

$$\int \frac{\mathrm{f}(x)}{\mathrm{g}(x)}\mathrm{d}x \neq \frac{\int \mathrm{f}(x)\ \mathrm{d}x}{\int \mathrm{g}(x)\ \mathrm{d}x}$$

1.3 Definite Integral

If f is a function **defined** and **continuous** on an interval $I = [u, v]$ and $\int \mathrm{f}(x)\,\mathrm{d}x = \mathrm{F}(x) + c$, then for a, $b \in I$ such that $u \le a < b \le v$, the ***definite integral*** of $\mathrm{f}(x)$ from a to b w.r.t. x is denoted by

$$\int_a^b \mathrm{f}(x)\,\mathrm{d}x = \big[\mathrm{F}(x)\big]_a^b = \mathrm{F}(b) - \mathrm{F}(a)$$

where ***a*** and ***b*** are called the ***lower*** and ***upper limits of the integral*** respectively.

1.3.1 Basic Properties of Definite Integrals

(i) $\displaystyle\int_a^a \mathrm{f}(x)\,\mathrm{d}x = 0$

(ii) $\displaystyle\int_a^b \mathrm{f}(x)\,\mathrm{d}x = -\int_b^a \mathrm{f}(x)\,\mathrm{d}x$

(iii) $\displaystyle\int_a^b \mathrm{f}(x)\,\mathrm{d}x = \int_a^c \mathrm{f}(x)\,\mathrm{d}x + \int_c^b \mathrm{f}(x)\,\mathrm{d}x$, where c is such that $a \le c \le b$

(iv) $\displaystyle\int_a^b \big[\mathrm{f}(x) \pm \mathrm{g}(x)\big]\mathrm{d}x = \int_a^b \mathrm{f}(x)\,\mathrm{d}x \pm \int_a^b \mathrm{g}(x)\,\mathrm{d}x$

(v) $\displaystyle\int_a^b k\,\mathrm{f}(x)\,\mathrm{d}x = k\int_a^b \mathrm{f}(x)\,\mathrm{d}x$, where k is any constant, $k \ne 0$

2 Integration of Standard Functions

2.1 Integration of the form x^n and Variations, where $n \ne -1$

Recall that $\dfrac{\mathrm{d}}{\mathrm{d}x}(x)^{n+1} = (n+1)x^n$ and $\dfrac{\mathrm{d}}{\mathrm{d}x}\big[\mathrm{f}(x)\big]^{n+1} = (n+1)\big[\mathrm{f}(x)\big]^n \mathrm{f}'(x)$.

For example,

$$\frac{\mathrm{d}}{\mathrm{d}x}(x)^5 = 5(x)^4$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(2x+5)^5 = 5(2x+5)^4(2) \qquad \mathrm{f}(x) = 2x+5, \quad \mathrm{f}'(x) = 2, \quad n = 4$$

$\displaystyle\int x^n \mathrm{d}x = \frac{x^{n+1}}{n+1} + c, \quad n \ne -1$	$\displaystyle\int \mathrm{f}'(x)\big[\mathrm{f}(x)\big]^n \mathrm{d}x = \frac{\big[\mathrm{f}(x)\big]^{n+1}}{n+1} + c, \quad n \ne -1$

Note: In particular, when $\mathrm{f}(x) = ax + b$, $a \in \mathbb{R} \setminus \{0\}$, $b \in \mathbb{R}$ are constants, then $\mathrm{f}'(x) = a$ and

$$\int a(ax+b)^n \,\mathrm{d}x = \frac{(ax+b)^{n+1}}{n+1} + c \quad \Rightarrow \int (ax+b)^n \,\mathrm{d}x = \frac{(ax+b)^{n+1}}{a(n+1)} + d, \quad n \ne -1$$

Example 1

Find

(a) $\int\left(\frac{1}{x^3}-\sqrt{x}\right)\mathrm{d}x$

$=\int\left(x^{-3}-x^{\frac{1}{2}}\right)\mathrm{d}x$

$=\frac{x^{-2}}{-2}-\frac{x^{\frac{3}{2}}}{\frac{3}{2}}+c=-\frac{1}{2x^2}-\frac{2}{3}x^{\frac{3}{2}}+c$

Check:

$\frac{\mathrm{d}}{\mathrm{d}x}\left(-\frac{1}{2x^2}-\frac{2}{3}x^{\frac{3}{2}}+c\right)=\frac{1}{x^3}-\sqrt{x}$

(b) $\int(5x+4)^2\,\mathrm{d}x$ $\quad \mathrm{f}(x)=5x+4$, $\mathrm{f}'(x)=5$, $n=2$

$=\frac{1}{5}\int 5(5x+4)^2\,\mathrm{d}x$

$=\frac{1}{5}\frac{(5x+4)^3}{3}+c$

$=\frac{1}{15}(5x+4)^3+c$

Check:

$\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{15}(5x+4)^3+c\right]=(5x+4)^3$

(c) $\int x\left(2x^2-3\right)^3\,\mathrm{d}x$ $\quad \mathrm{f}(x)=2x^2-3$, $\mathrm{f}'(x)=4x$, $n=3$

$=\frac{1}{4}\int 4x\left(2x^2-3\right)^3\,\mathrm{d}x$

$=\frac{1}{4}\frac{\left(2x^2-3\right)^4}{4}+c$

$=\frac{1}{16}\left(2x^2-3\right)^4+c$

Check:

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{16}\left(2x^2-3\right)^4+c\right)=x(2x^2-3)^3$

(d) $\int\frac{2x+3}{(x^2+3x)^4}\,\mathrm{d}x$ $\quad \mathrm{f}(x)=x^2+3x$, $\mathrm{f}'(x)=2x+3$, $n=-4$

$=\int(2x+3)\left(x^2+3x\right)^{-4}\,\mathrm{d}x$

$=-\frac{1}{3\left(x^2+3x\right)^3}+c$

Check:

$\frac{\mathrm{d}}{\mathrm{d}x}\left(-\frac{1}{3\left(x^2+3x\right)^3}+c\right)=\frac{2x+3}{(x^2+3x)^4}$

2.2 Integration of the form x^{-1} and Variations

$$\ln|x|=\begin{cases}\ln x, & x>0\\ \ln(-x), & x<0,\end{cases}\qquad \frac{\mathrm{d}}{\mathrm{d}x}\ln|x|=\begin{cases}\frac{\mathrm{d}}{\mathrm{d}x}\ln x=\frac{1}{x}, & x>0\\ \frac{\mathrm{d}}{\mathrm{d}x}\ln(-x)=\frac{-1}{-x}=\frac{1}{x}, & x<0.\end{cases}$$

When the domain is not specified, we write $\int\frac{1}{x}\,\mathrm{d}x = \ln|x|+c$.

In general, we have $\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)}\mathrm{d}x = \ln\left|\mathrm{f}(x)\right| + c$.

$\int \frac{1}{x}\mathrm{d}x = \ln\lvert x\rvert + c$	$\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)}\mathrm{d}x = \ln\left\lvert\mathrm{f}(x)\right\rvert + c$

Example 2

Find

(a)

$$\int \frac{3}{x}\,\mathrm{d}x$$
$$= 3\int \frac{1}{x}\,\mathrm{d}x$$
$$= 3\ln|x| + c$$

(b) $\mathrm{f}(x) = 1-5x$, $\mathrm{f}'(x) = -5$

$$\int \frac{1}{1-5x}\,\mathrm{d}x$$
$$= -\frac{1}{5}\int \frac{-5}{1-5x}\,\mathrm{d}x$$
$$= -\frac{1}{5}\ln|1-5x| + c$$

(c) $\mathrm{f}(x) = 2x^2 - 2x + 1$, $\mathrm{f}'(x) = 4x - 2$

$$\int \frac{2x-1}{\left(2x^2-2x+1\right)}\,\mathrm{d}x$$
$$= \frac{1}{2}\int \frac{4x-2}{\left(2x^2-2x+1\right)}\,\mathrm{d}x$$
$$= \frac{1}{2}\ln\left|2x^2-2x+1\right| + c$$

(d) $\mathrm{f}(x) = \mathrm{e}^x - 2$, $\mathrm{f}'(x) = \mathrm{e}^x$

$$\int \frac{1}{1-2\mathrm{e}^{-x}}\,\mathrm{d}x$$
$$= \int \frac{\mathrm{e}^x}{\mathrm{e}^x-2}\,\mathrm{d}x$$
$$= \ln\left|\mathrm{e}^x-2\right| + c$$

2.3 Integration of Exponential Function and Variations

Since $\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^x\right) = \mathrm{e}^x$ and $\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\mathrm{f}(x)}\right) = \mathrm{f}'(x)\mathrm{e}^{\mathrm{f}(x)}$, we have

$\int \mathrm{e}^x\,\mathrm{d}x = \mathrm{e}^x + c$	$\int \mathrm{f}'(x)\,\mathrm{e}^{\mathrm{f}(x)}\,\mathrm{d}x = \mathrm{e}^{\mathrm{f}(x)} + c$

Example 3

Find

(a) $\mathrm{f}(x) = 3x+2$, $\mathrm{f}'(x) = 3$

$$\int_0^1 \mathrm{e}^{3x+2}\mathrm{d}x$$
$$= \frac{1}{3}\int_0^1 3\mathrm{e}^{3x+2}\mathrm{d}x$$
$$= \left[\frac{1}{3}\mathrm{e}^{3x+2}\right]_0^1 = \frac{1}{3}\left(\mathrm{e}^5 - \mathrm{e}^2\right)$$

(b) $\mathrm{f}(x) = x^2$, $\mathrm{f}'(x) = 2x$

$$\int x\mathrm{e}^{x^2}\,\mathrm{d}x$$
$$= \frac{1}{2}\int (2x)\,\mathrm{e}^{x^2}\,\mathrm{d}x$$
$$= \frac{1}{2}\mathrm{e}^{x^2} + c$$

(c) $\int 4^{3x+2}\,\mathrm{d}x$

$= \int \mathrm{e}^{(3x+2)\ln 4}\,\mathrm{d}x$

$= \dfrac{1}{3\ln 4}\mathrm{e}^{(3x+2)\ln 4} + c$

$= \dfrac{1}{3\ln 4}4^{3x+2} + c$

*note: we can write a^x as $\mathrm{e}^{\ln a^x}$

(d) $\int (\sin x + x\cos x)\mathrm{e}^{x\sin x}$, $\mathrm{f}(x) = x\sin x$

$= \mathrm{e}^{x\sin x} + c$ $\mathrm{f}'(x) = \sin x + x\cos x$

2.4 Integration of Standard Trigonometric Functions and Variations

$\int \cos x\,\mathrm{d}x = \sin x + c$	$\int \mathrm{f}'(x)\cos\big(\mathrm{f}(x)\big)\,\mathrm{d}x = \sin\big(\mathrm{f}(x)\big) + c$
$\int \sin x\,\mathrm{d}x = -\cos x + c$	$\int \mathrm{f}'(x)\sin\big(\mathrm{f}(x)\big)\,\mathrm{d}x = -\cos\big(\mathrm{f}(x)\big) + c$
$\int \sec^2 x\,\mathrm{d}x = \tan x + c$	$\int \mathrm{f}'(x)\sec^2\big(\mathrm{f}(x)\big)\,\mathrm{d}x = \tan\big(\mathrm{f}(x)\big) + c$

Example 4

Find

(a) $\int (3\sin x + \cos 4x)\,\mathrm{d}x$

$= -3\cos x + \dfrac{1}{4}\sin 4x + c$

(b) $\int 2\sin(1-2x)\,\mathrm{d}x$

$= -\int -2\sin(1-2x)\,\mathrm{d}x$

$= \cos(1-2x) + c$

(c) $\int 3\sec^2(5x-2)\,\mathrm{d}x$

$= \dfrac{3}{5}\int 5\sec^2(5x-2)\,\mathrm{d}x$

$= \dfrac{3}{5}\tan(5x-2) + c$

(d) $\int x\cos(3x^2)\,\mathrm{d}x$

$= \dfrac{1}{6}\int 6x\cos(3x^2)\,\mathrm{d}x$

$= \dfrac{1}{6}\sin(3x^2) + c$

2.4.1 Integration using Trigonometric Identities

Commonly used trigonometry identities (to be used for simplification)

- Basic trigonometric identities
  - $\sin^2 x + \cos^2 x = 1$
  - $\tan^2 x + 1 = \sec^2 x$
  - $\cot^2 x + 1 = \operatorname{cosec}^2 x$

- Double angle formula:
  - $\sin 2x = 2\sin x \cos x$
  - $\cos 2x = 2\cos^2 x - 1 = 1 - 2\sin^2 x = \cos^2 x - \sin^2 x$

Example 5

Find

(a) $\int \tan^2 x \, \mathrm{d}x$

$= \int \left(\sec^2 x - 1\right) \, \mathrm{d}x$

$= \tan x - x + c$

(b) $\int \cos^2 x \, \mathrm{d}x$

$= \frac{1}{2}\int \left(1 + \cos 2x\right) \, \mathrm{d}x$

$= \frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right) + c$

(c) $\int \sin 3x \cos 3x \, \mathrm{d}x$

$= \frac{1}{2}\int 2\sin 3x \cos 3x \, \mathrm{d}x$

$= \frac{1}{2(6)}\int 6\sin 6x \, \mathrm{d}x$

$= -\frac{1}{12}\cos 6x + c$

(d) $\int \frac{1}{1+\cos 2x} \, \mathrm{d}x$

$= \int \frac{1}{1+\left(2\cos^2 x - 1\right)} \, \mathrm{d}x$

$= \frac{1}{2}\int \sec^2 x \, \mathrm{d}x$

$= \frac{1}{2}\tan x + c$

Sometimes**,** we may also need to apply the results in sections **2.1** and **2.2**; namely

$$\int \mathrm{f}'(x)\left[\mathrm{f}(x)\right]^n \mathrm{d}x = \frac{\left[\mathrm{f}(x)\right]^{n+1}}{n+1} + c, \quad n \neq -1 \text{ and } \int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \mathrm{d}x = \ln\left|\mathrm{f}(x)\right| + c.$$

Example 6

Find

(a)

$$\int \frac{\sec^2\theta}{1+\tan\theta}\,\mathrm{d}\theta$$
$$= \ln\left|1+\tan\theta\right| + c$$

(b)

$$\int \sec x\ \mathrm{d}x$$
$$= \int \sec x \times \left(\frac{\sec x + \tan x}{\sec x + \tan x}\right) \mathrm{d}x$$
$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\ \mathrm{d}x$$
$$= \ln\left|\sec x + \tan x\right| + c$$

(c)

$$\int \tan x\ \mathrm{d}x$$
$$= -\int \frac{-\sin x}{\cos x}\,\mathrm{d}x$$
$$= -\ln\left|\cos x\right| + c$$
$$= \ln\left|\frac{1}{\cos x}\right| + c$$
$$= \ln\left|\cos x\right|^{-1} + c$$
$$= \ln\left|\sec x\right| + c$$

(d)

$$\int \sin x \cos^2 x\,\mathrm{d}x$$
$$= -\int (-\sin x)(\cos x)^2\ \mathrm{d}x$$
$$= -\frac{\cos^3 x}{3} + c$$

3 Other Integration Techniques

3.1 Integration of $\frac{1}{\sqrt{a^2-x^2}}$, $\frac{1}{a^2+x^2}$ ($a > 0$)

Recall that $\frac{\mathrm{d}}{\mathrm{d}x}\left(\sin^{-1}x\right)=\frac{1}{\sqrt{1-x^2}}$ and $\frac{\mathrm{d}}{\mathrm{d}x}\left(\tan^{-1}x\right)=\frac{1}{1+x^2}$.

Therefore, $\int\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x=\sin^{-1}x+c$ and $\int\frac{1}{1+x^2}\,\mathrm{d}x=\tan^{-1}x+c$.

This result can be extended to find $\int\frac{1}{\sqrt{a^2-x^2}}\,\mathrm{d}x$ and $\int\frac{1}{a^2+x^2}\,\mathrm{d}x$.

$\int\frac{1}{\sqrt{a^2-x^2}}\,\mathrm{d}x=\sin^{-1}\frac{x}{a}+c$, $\lvert x\rvert<a$	$\int\frac{\mathrm{f}'(x)}{\sqrt{a^2-\left[\mathrm{f}(x)\right]^2}}\,\mathrm{d}x=\sin^{-1}\frac{\mathrm{f}(x)}{a}+c$
$\int\frac{1}{a^2+x^2}\,\mathrm{d}x=\frac{1}{a}\tan^{-1}\frac{x}{a}+c$	$\int\frac{\mathrm{f}'(x)}{a^2+\left[\mathrm{f}(x)\right]^2}\,\mathrm{d}x=\frac{1}{a}\tan^{-1}\frac{\mathrm{f}(x)}{a}+c$

Example 7

Find

(a)

$$\int\frac{1}{\sqrt{4-x^2}}\,\mathrm{d}x$$
$$=\int\frac{1}{\sqrt{2^2-x^2}}\,\mathrm{d}x$$
$$=\sin^{-1}\left(\frac{x}{2}\right)+c$$

(b)

$$\int\frac{2}{3+x^2}\,\mathrm{d}x$$
$$=2\int\frac{1}{\left(\sqrt{3}\right)^2+x^2}\,\mathrm{d}x$$
$$=\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{x}{\sqrt{3}}\right)+c$$

(c)

$$\int\frac{1}{(2x+1)^2+9}\,\mathrm{d}x$$
$$=\frac{1}{2}\int\frac{2}{(2x+1)^2+3^2}\,\mathrm{d}x \qquad \mathrm{f}(x)=2x+1,\ \mathrm{f}'(x)=2$$
$$=\frac{1}{6}\tan^{-1}\left(\frac{2x+1}{3}\right)+c$$

(d)

$$\int\frac{1}{x^2+2x+5}\,\mathrm{d}x$$
$$=\int\frac{1}{(x+1)^2+2^2}\,\mathrm{d}x \qquad \mathrm{f}(x)=x+1,\ \mathrm{f}'(x)=1$$
$$=\frac{1}{2}\tan^{-1}\left(\frac{x+1}{2}\right)+c$$

(e)

$$\int \frac{1}{\sqrt{-12+16x-4x^2}}\,\mathrm{d}x$$
$$= \int \frac{1}{2\sqrt{-x^2+4x-3}}\,\mathrm{d}x$$
$$= \int \frac{1}{2\sqrt{-\left(x^2-4x+3\right)}}\,\mathrm{d}x$$
$$= \int \frac{1}{2\sqrt{-\left(x^2-4x+4-1\right)}}\,\mathrm{d}x$$
$$= \int \frac{1}{2\sqrt{1-(x-2)^2}}\,\mathrm{d}x \qquad \mathrm{f}(x)=x-2,\ \mathrm{f}'(x)=1$$
$$= \frac{1}{2}\sin^{-1}(x-2)+c$$

(f)

$$\int \frac{x}{\sqrt{1-x^4}}\,\mathrm{d}x$$
$$= \frac{1}{2}\int \frac{2x}{\sqrt{1-\left(x^2\right)^2}}\,\mathrm{d}x \qquad \mathrm{f}(x)=x^2,\ \mathrm{f}'(x)=2x$$
$$= \frac{1}{2}\sin^{-1}\left(x^2\right)+c$$

3.2 Integration Using Partial Fractions

Consider the integral $\int \frac{3x+4}{(x+2)(x+1)}\,\mathrm{d}x$.

By partial fractions, $\frac{3x+4}{(x+2)(x+1)} = \frac{2}{x+2}+\frac{1}{x+1}$

Thus

$$\int \frac{3x+4}{(x+2)(x+1)}\,\mathrm{d}x = \int\left[\frac{2}{x+2}+\frac{1}{x+1}\right]\mathrm{d}x$$
$$= 2\ln|x+2|+\ln|x+1|+c$$

The following formulae for decomposition of partial fractions are given in MF 27.

Non-repeated linear factors:

$$\frac{px+q}{(ax+b)(cx+d)} = \frac{A}{(ax+b)}+\frac{B}{(cx+d)}$$

Repeated linear factors:

$$\frac{px^2+qx+r}{(ax+b)(cx+d)^2} = \frac{A}{(ax+b)}+\frac{B}{(cx+d)}+\frac{C}{(cx+d)^2}$$

Non-repeated quadratic factor:

$$\frac{px^2+qx+r}{(ax+b)\left(x^2+c^2\right)} = \frac{A}{(ax+b)}+\frac{Bx+C}{\left(x^2+c^2\right)}$$

Example 8

Find

(a)	**For non-repeated linear factors:** $\int \frac{1}{(x+1)(2x-1)} \mathrm{d}x$ $= \int \left[-\frac{1}{3(x+1)} + \frac{2}{3(2x-1)} \right] \mathrm{d}x$ $= -\frac{1}{3}\ln\lvert x+1\rvert + \frac{1}{3}\ln\lvert 2x-1\rvert + c$	Let $\frac{1}{(x+1)(2x-1)} = \frac{A}{x+1} + \frac{B}{2x-1}$ $1 = A(2x-1) + B(x+1)$ When $x=-1$, $A = -\frac{1}{3}$ When $x=\frac{1}{2}$, $B = \frac{2}{3}$ $\therefore \frac{1}{(x+1)(2x-1)} = -\frac{1}{3(x+1)} + \frac{2}{3(2x-1)}$
(b)	**For repeated linear factors:** $\int \frac{1}{(x+1)(x+2)^2} \mathrm{d}x$ $= \int \left[\frac{1}{x+1} - \frac{1}{x+2} - \frac{1}{(x+2)^2} \right] \mathrm{d}x$ $= \ln\lvert x+1\rvert - \ln\lvert x+2\rvert + \frac{1}{x+2} + c$	Let $\frac{1}{(x+1)(x+2)^2} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$ $1 = A(x+2)^2 + B(x+1)(x+2) + C(x+1)$ When $x=-1$, $A=1$ When $x=-2$, $C=-1$ By comparing coefficient of x^2, $0 = A+B$, so $B=-1$. $\therefore \frac{1}{(x+1)(x+2)^2} = \frac{1}{x+1} - \frac{1}{x+2} - \frac{1}{(x+2)^2}$

(c) **For non-repeated quadratic factors:**

$$\int \frac{5}{(x+1)(x^2+4)}\,\mathrm{d}x$$

$$= \int \left[\frac{1}{(x+1)} + \frac{-x+1}{(x^2+4)}\right]\mathrm{d}x$$

$$= \int \left[\frac{1}{(x+1)} + \frac{-x}{(x^2+4)} + \frac{1}{(x^2+4)}\right]\mathrm{d}x$$

$$= \ln|x+1| - \frac{1}{2}\int \frac{2x}{x^2+4}\,\mathrm{d}x + \frac{1}{2}\tan^{-1}\frac{x}{2}$$

$$= \ln|x+1| - \frac{1}{2}\ln\left(x^2+4\right) + \frac{1}{2}\tan^{-1}\frac{x}{2} + c$$

Let $\dfrac{5}{(x+1)(x^2+4)} = \dfrac{A}{x+1} + \dfrac{Bx+C}{x^2+4}$

$5 = A(x^2+4) + (Bx+C)(x+1)$

When $x=-1$, $A=1$

When $x=0$, $C=1$

By comparing coefficient of x,

$0 = B+1 \quad \Rightarrow \quad B=-1$

$$\frac{5}{(x+1)(x^2+4)} = \frac{1}{(x+1)} + \frac{-x+1}{(x^2+4)}$$

Example 9 Find $\int \dfrac{2x^2-7x-1}{(x+1)(x-2)}\,\mathrm{d}x$.

$$\int \frac{2x^2-7x-1}{(x+1)(x-2)}\,\mathrm{d}x$$

$$= \int 2 - \frac{8}{3(x+1)} - \frac{7}{3(x-2)}\,\mathrm{d}x$$

$$= 2x - \frac{8}{3}\ln|x+1| - \frac{7}{3}\ln|x-2| + c$$

Let $\dfrac{2x^2-7x-1}{(x+1)(x-2)} = A + \dfrac{B}{(x+1)} + \dfrac{C}{(x-2)}$

By observation, $A=2$

$2x^2-7x-1 = 2(x+1)(x-2) + B(x-2) + C(x+1)$

When $x=-1$, $B=-\dfrac{8}{3}$

When $x=2$, $C=-\dfrac{7}{3}$

$$\frac{2x^2-7x-1}{(x+1)(x-2)} = 2 - \frac{8}{3(x+1)} - \frac{7}{3(x-2)}$$

3.2.1 Integration of $\frac{1}{x^2-a^2}$ and $\frac{1}{a^2-x^2}$ $(a>0)$

Using partial fraction, $\frac{1}{x^2-a^2}=\frac{1}{2a}\left(\frac{1}{x-a}-\frac{1}{x+a}\right)$. Hence

$$\int \frac{1}{x^2-a^2}\,\mathrm{d}x=\int \frac{1}{2a}\left(\frac{1}{x-a}-\frac{1}{x+a}\right)\mathrm{d}x=\frac{1}{2a}\left(\ln|x-a|-\ln|x+a|\right)+c=\frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|+c.$$

Hence we have the following formulae.

$\int \frac{1}{x^2-a^2}\,\mathrm{d}x=\frac{1}{2a}\ln\left(\frac{x-a}{x+a}\right)+c,\ x>a$	$\int \frac{1}{a^2-x^2}\,\mathrm{d}x=\frac{1}{2a}\ln\left(\frac{a+x}{a-x}\right)+c,\ \lvert x\rvert<a$

Note : For integrals similar to the forms $\int \frac{1}{x^2-a^2}\,\mathrm{d}x$ and $\int \frac{1}{a^2-x^2}\,\mathrm{d}x$ $(a>0)$, the above result may be quoted without proof.

In general, $$\int \frac{\mathrm{f}'(x)}{[\mathrm{f}(x)]^2-a^2}\,\mathrm{d}x=\frac{1}{2a}\ln\left|\frac{\mathrm{f}(x)-a}{\mathrm{f}(x)+a}\right|+c \text{ and}$$

$$\int \frac{\mathrm{f}'(x)}{a^2-[\mathrm{f}(x)]^2}\,\mathrm{d}x=\frac{1}{2a}\ln\left|\frac{a+\mathrm{f}(x)}{a-\mathrm{f}(x)}\right|+c.$$

Example 10

Find

(a)

$$\int \frac{1}{2x^2-4}\,\mathrm{d}x$$
$$=\frac{1}{2}\int \frac{1}{x^2-2}\,\mathrm{d}x$$
$$=\frac{1}{2}\int \frac{1}{x^2-\left(\sqrt{2}\right)^2}\,\mathrm{d}x$$
$$=\frac{1}{2}\times\frac{1}{2(\sqrt{2})}\ln\left|\frac{x-\sqrt{2}}{x+\sqrt{2}}\right|+c$$
$$=\frac{1}{4\sqrt{2}}\ln\left|\frac{x-\sqrt{2}}{x+\sqrt{2}}\right|+c$$

(b)

$$\int \frac{1}{x^2+4x-5}\,\mathrm{d}x$$
$$=\int \frac{1}{x^2+4x+4-9}\,\mathrm{d}x$$
$$=\int \frac{1}{(x+2)^2-3^2}\,\mathrm{d}x$$
$$=\frac{1}{2(3)}\ln\left|\frac{x+2-3}{x+2+3}\right|+c$$
$$=\frac{1}{6}\ln\left|\frac{x-1}{x+5}\right|+c$$

Example 11

By considering $\frac{\mathrm{d}}{\mathrm{d}x}(4x^2+4x-8)$, find $\int \frac{8x+5}{4x^2+4x-8}\,\mathrm{d}x$.

$$\frac{\mathrm{d}}{\mathrm{d}x}(4x^2+4x-8)=8x+4$$

$$\begin{aligned}\int \frac{8x+5}{4x^2+4x-8}\,\mathrm{d}x &= \int \frac{8x+4+1}{4x^2+4x-8}\mathrm{d}x \\ &= \int \frac{8x+4}{4x^2+4x-8}+\frac{1}{4x^2+4x-8}\mathrm{d}x \\ &= \ln\left|4x^2+4x-8\right|+\frac{1}{2}\int \frac{2}{(2x+1)^2-3^2}\mathrm{d}x \\ &= \ln\left|4x^2+4x-8\right|+\frac{1}{12}\ln\left|\frac{2x+1-3}{2x+1+3}\right|+c \\ &= \ln\left|4x^2+4x-8\right|+\frac{1}{12}\ln\left|\frac{x-1}{x+2}\right|+c\end{aligned}$$

3.3 Integration by Substitution (Change of Variable)

Consider the integral $\int \frac{(\ln x)^3}{x}\,\mathrm{d}x$.

Here we introduce "something extra" and this "something extra" is a new variable. We transform the integral involving the variable x to an integral involving a new variable, say u. The idea is illustrated below.

Let $u=\ln x$, then $\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{x} \Rightarrow \mathrm{d}u=\frac{1}{x}\mathrm{d}x$

$$\begin{aligned}&\therefore \int \frac{(\ln x)^3}{x}\,\mathrm{d}x \\ &= \int (\ln x)^3 \cdot \frac{1}{x}\,\mathrm{d}x \\ &= \int u^3\,\mathrm{d}u \\ &= \frac{1}{4}u^4+c \\ &= \frac{(\ln x)^4}{4}+c\end{aligned}$$

In general, this method works when we can write the integral in the form $\int \mathrm{f}(\mathrm{g}(x))\mathrm{g}'(x)\,\mathrm{d}x$.

In the example above, $\mathrm{f}(x)=x^3$, $\mathrm{g}(x)=\ln x$.

Substitution Rule

$$\int \mathrm{f}(\mathrm{g}(x))\mathrm{g}'(x)\,\mathrm{d}x = \int \mathrm{f}(u)\,\mathrm{d}u \qquad \text{where } u = \mathrm{g}(x)$$

Step 1: Substitute $u = \mathrm{g}(x)$ and $\mathrm{d}u = \mathrm{g}'(x)\,\mathrm{d}x$ to obtain the integral $\int \mathrm{f}(u)\,\mathrm{d}u$

Step 2: Integrate with respect to u

Step 3: Replace u by $\mathrm{g}(x)$ in the result

For definite integrals,

$$\int_a^b \mathrm{f}(\mathrm{g}(x))\mathrm{g}'(x)\,\mathrm{d}x = \int_{\mathrm{g}(a)}^{\mathrm{g}(b)} \mathrm{f}(u)\,\mathrm{d}u \qquad \text{where } u = \mathrm{g}(x)$$

Example 12

Using the given substitution, find

(a)

$$\int \frac{\mathrm{e}^{\sqrt{x}}}{\sqrt{x}}\,\mathrm{d}x$$

$$= 2\int \mathrm{e}^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}\,\mathrm{d}x$$

$$= 2\int \mathrm{e}^{u}\,\mathrm{d}u$$

$$= 2\mathrm{e}^{u} + c$$

$$= 2\mathrm{e}^{\sqrt{x}} + c$$

Remember to substitute back the original variable for an indefinite integral

Let $u = \sqrt{x}$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2\sqrt{x}} \Rightarrow \mathrm{d}u = \frac{1}{2\sqrt{x}}\mathrm{d}x$$

(b)

$$\int \frac{\mathrm{e}^{x}}{\mathrm{e}^{x} + \mathrm{e}^{-x}}\,\mathrm{d}x$$

$$= \int \frac{1}{\mathrm{e}^{x} + \mathrm{e}^{-x}} \cdot \mathrm{e}^{x}\mathrm{d}x$$

$$= \int \frac{1}{u + u^{-1}}\,\mathrm{d}u$$

$$= \int \frac{u}{u^2 + 1}\,\mathrm{d}u$$

$$= \frac{1}{2}\int \frac{2u}{u^2 + 1}\,\mathrm{d}u$$

$$= \frac{1}{2}\ln(u^2 + 1) + c$$

$$= \frac{1}{2}\ln(\mathrm{e}^{2x} + 1) + c$$

Let $u = \mathrm{e}^{x}$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^{x} \Rightarrow \mathrm{d}u = \mathrm{e}^{x}\mathrm{d}x$$

(c)	$\int x\sqrt{x-2}\ \mathrm{d}x$ $= \int \left(u^2+2\right)(u)\ 2u\ \mathrm{d}u$ $= 2\int u^4 + 2u^2\ \mathrm{d}u$ $= 2\left[\frac{u^5}{5}+\frac{2u^3}{3}\right]+c$ $= \frac{2\sqrt{(x-2)^5}}{5}+\frac{4\sqrt{(x-2)^3}}{3}+c$	Let $u=\sqrt{x-2}$ $\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{2\sqrt{x-2}} \Rightarrow 2u\,\mathrm{d}u=\mathrm{d}x$ $u=\sqrt{x-2} \Rightarrow x=u^2+2$

[Do refer to your secondary school notes on how definite integrals are evaluated.]

Example 13

Using the given substitution, evaluate the following definite integrals, leaving your answer in the exact form.

(a)	$\int_{\frac{2}{\sqrt{3}}}^{2} \frac{1}{x\sqrt{x^2-1}}\mathrm{d}x \qquad (\text{Substitute } x=\frac{1}{y})$ $= \int_{\frac{\sqrt{3}}{2}}^{\frac{1}{2}} \frac{y}{\sqrt{\frac{1}{y^2}-1}}\left(-\frac{1}{y^2}\right)\mathrm{d}y$ $= \int_{\frac{\sqrt{3}}{2}}^{\frac{1}{2}} \frac{y^2}{\sqrt{1-y^2}}\left(-\frac{1}{y^2}\right)\mathrm{d}y$ $= -\int_{\frac{\sqrt{3}}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{1-y^2}}\mathrm{d}y$ $= -\left[\ \sin^{-1}y\ \right]_{\frac{\sqrt{3}}{2}}^{\frac{1}{2}}$ $= -\left[\ \sin^{-1}\left(\frac{1}{2}\right)-\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\ \right]$ $= -\left(\frac{\pi}{6}-\frac{\pi}{3}\right) = \frac{\pi}{6}$	Let $x=\frac{1}{y}$ $\frac{\mathrm{d}x}{\mathrm{d}y}=-\frac{1}{y^2} \Rightarrow \mathrm{d}x=-\frac{1}{y^2}\mathrm{d}y$ When $x=2,\ y=\frac{1}{2}$ When $x=\frac{2}{\sqrt{3}},\ y=\frac{\sqrt{3}}{2}$
(b)	$\int_{2}^{3} \frac{1}{\sqrt{(3-x)(x-2)}}\mathrm{d}x \quad (\text{Substitute } x=2\cos^2\theta+3\sin^2\theta)$ $= \int_{0}^{\frac{\pi}{2}} \frac{1}{\sqrt{(1-\sin^2\theta)(\sin^2\theta)}}(2\sin\theta\cos\theta)\ \mathrm{d}\theta$ $= \int_{0}^{\frac{\pi}{2}} \frac{1}{\cos\theta\sin\theta}(2\sin\theta\cos\theta)\ \mathrm{d}\theta$ $= \int_{0}^{\frac{\pi}{2}} 2\ \mathrm{d}\theta = \left[2\theta\right]_0^{\frac{\pi}{2}} = \pi$	$x=2\cos^2\theta+3\sin^2\theta$ $\quad =2+\sin^2\theta$ $\frac{\mathrm{d}x}{\mathrm{d}\theta}=2\sin\theta\cos\theta$ $\Rightarrow \mathrm{d}x=2\sin\theta\cos\theta\,\mathrm{d}\theta$ When $x=2,\ \theta=0$ When $x=3,\ \theta=\frac{\pi}{2}$

(c)

$$\int_0^{\frac{1}{2}} \frac{1}{\left(1-x^2\right)^{\frac{3}{2}}}\,\mathrm{d}x$$

$$\int_0^{\frac{1}{2}} \frac{1}{\left(1-x^2\right)^{\frac{3}{2}}}\,\mathrm{d}x$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{(1-\sin^2\theta)^{\frac{3}{2}}}\cos\theta\,\mathrm{d}\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{\cos^3\theta}\cos\theta\,\mathrm{d}\theta$$

$$= \int_0^{\frac{\pi}{6}} \sec^2\theta\,\mathrm{d}\theta$$

$$= \left[\tan\theta\right]_0^{\frac{\pi}{6}} = \left[\tan\left(\tfrac{\pi}{6}\right)-\tan(0)\right] = \frac{1}{\sqrt{3}}$$

Let $x = \sin\theta$

$\dfrac{\mathrm{d}x}{\mathrm{d}\theta} = \cos\theta \Rightarrow \mathrm{d}x = \cos\theta\,\mathrm{d}\theta$

When $x = \dfrac{1}{2}$, $\theta = \dfrac{\pi}{6}$

When $x = 0$, $\theta = 0$

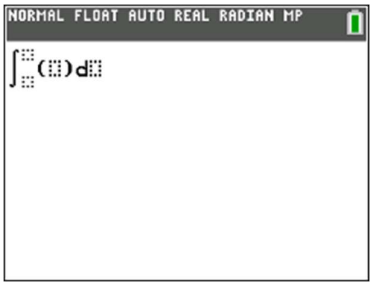
For questions involving definite integrals, you can verify the accuracy of your answers using the numerical integration function of the GC.

With reference to Example 13(c).

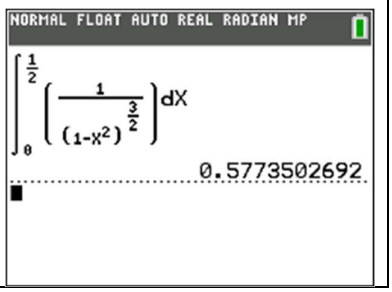
1. Press [ALPHA] [WINDOW] and select 4



2. Key in $\displaystyle\int_0^{\frac{1}{2}} \frac{1}{(1-x^2)^{\frac{3}{2}}}\,\mathrm{d}x$



3. Press ENTER , you would obtain the numerical answer 0.5773502692



3.4 Integration by Parts

This method is usually used when the integrands are products of two distinct types of functions such as $\ln x$, e^x, $\sin x$ etc. It is based on the product rule of differentiation, and its purpose is to express the integral of a product in terms of a second integral that is usually easier to find.

From the product rule of differentiation, we have

$$\frac{\mathrm{d}}{\mathrm{d}x}(uv) = u\frac{\mathrm{d}v}{\mathrm{d}x} + v\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Rewriting, we get

$$u\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}(uv) - v\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Integrating with respect to x, we obtain

$$\boxed{\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x}$$

To use this formula we need to choose appropriate u and $\frac{\mathrm{d}v}{\mathrm{d}x}$, and find $\frac{\mathrm{d}u}{\mathrm{d}x}$ and v.

Consider the following table and compare the two ways of assigning u and $\frac{\mathrm{d}v}{\mathrm{d}x}$ for $\int x\mathrm{e}^{2x}\,\mathrm{d}x$.

	u	$\frac{\mathrm{d}v}{\mathrm{d}x}$	v	$\frac{\mathrm{d}u}{\mathrm{d}x}$	Integration by parts formula
Option 1	x	e^{2x}	$\frac{1}{2}\mathrm{e}^{2x}$	1	$\int x\mathrm{e}^{2x}\,\mathrm{d}x = \frac{1}{2}x\mathrm{e}^{2x} - \int \frac{1}{2}\mathrm{e}^{2x}\,\mathrm{d}x$
Option 2	e^{2x}	x	$\frac{1}{2}x^2$	$2\mathrm{e}^{2x}$	$\int x\mathrm{e}^{2x}\mathrm{d}x = \frac{1}{2}x^2\mathrm{e}^{2x} - \int x^2\mathrm{e}^{2x}\mathrm{d}x$

The first option gives rise to $\int \frac{1}{2}\mathrm{e}^{2x}\,\mathrm{d}x$, which can be found easily.

The second option leads to $\int x^2\mathrm{e}^{2x}\,\mathrm{d}x$, which is even harder to find than $\int x\mathrm{e}^{2x}\,\mathrm{d}x$.

Hence, using the first option,

$$\begin{aligned}\int x\mathrm{e}^{2x}\,\mathrm{d}x &= \frac{1}{2}x\mathrm{e}^{2x} - \int \frac{1}{2}\mathrm{e}^{2x}\,\mathrm{d}x \\ &= \frac{1}{2}x\mathrm{e}^{2x} - \frac{1}{4}\mathrm{e}^{2x} + c.\end{aligned}$$

Example 14

Find

(a)

$$\int x\sin x\ \mathrm{d}x$$

$$= x(-\cos x) - \int (1)(-\cos x)\ \mathrm{d}x$$

$$= -x\cos x + \sin x + c$$

$u = x$	$\frac{\mathrm{d}v}{\mathrm{d}x} = \sin x$
$\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ ←	$v = -\cos x$

(b)

$$\int_0^1 \tan^{-1} x\ \mathrm{d}x$$

$$= \int_0^1 (1)\left(\tan^{-1} x\right)\mathrm{d}x$$

$$= \left[x\tan^{-1} x\right]_0^1 - \int_0^1 \frac{x}{1+x^2}\mathrm{d}x$$

$$= \left[x\tan^{-1} x - \frac{1}{2}\ln\left(1+x^2\right)\right]_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2}\ln 2$$

$u = \tan^{-1} x$	$\frac{\mathrm{d}v}{\mathrm{d}x} = 1$
$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x^2}$ ←	$v = x$

Important Technique:
Insert "1" so as to make the integrand a product of two terms.

(c)

$$\int x^2 \mathrm{e}^{2x}\ \mathrm{d}x$$

$$= \frac{1}{2}x^2\mathrm{e}^{2x} - \int (2x)\left(\frac{1}{2}\mathrm{e}^{2x}\right)\mathrm{d}x$$

$$= \frac{1}{2}x^2\mathrm{e}^{2x} - \int x\mathrm{e}^{2x}\ \mathrm{d}x$$

$$= \frac{1}{2}x^2\mathrm{e}^{2x} - \left(\frac{1}{2}x\mathrm{e}^{2x} - \int (1)\left(\frac{1}{2}\mathrm{e}^{2x}\right)\mathrm{d}x\right)$$

$$= \frac{1}{2}x^2\mathrm{e}^{2x} - \frac{1}{2}x\mathrm{e}^{2x} + \int \frac{1}{2}\mathrm{e}^{2x}\ \mathrm{d}x$$

$$= \frac{1}{2}x^2\mathrm{e}^{2x} - \frac{1}{2}x\mathrm{e}^{2x} + \frac{1}{4}\mathrm{e}^{2x} + c$$

$u = x^2$	$\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{2x}$
$\frac{\mathrm{d}u}{\mathrm{d}x} = 2x$ ←	$v = \frac{1}{2}\mathrm{e}^{2x}$

$u = x$	$\frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{2x}$
$\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ ←	$v = \frac{1}{2}\mathrm{e}^{2x}$

(d)

$$\int x^3\mathrm{e}^{x^2}\ \mathrm{d}x$$

$$= \int \left(x^2\right)\left(x\mathrm{e}^{x^2}\right)\mathrm{d}x$$

$$= \frac{1}{2}x^2\mathrm{e}^{x^2} - \int (2x)\left(\frac{1}{2}\mathrm{e}^{x^2}\right)\mathrm{d}x$$

$$= \frac{1}{2}x^2\mathrm{e}^{x^2} - \frac{1}{2}\mathrm{e}^{x^2} + c$$

$u = x^2$	$\frac{\mathrm{d}v}{\mathrm{d}x} = x\mathrm{e}^{x^2}$
$\frac{\mathrm{d}u}{\mathrm{d}x} = 2x$ ←	$v = \frac{1}{2}\mathrm{e}^{x^2}$

(e)

$$\int e^x \cos x \, dx$$

$$= e^x \cos x - \int e^x (-\sin x) \, dx$$

$$= e^x \cos x + \int e^x (\sin x) \, dx$$

$$= e^x \cos x + \left[e^x \sin x - \int e^x \cos x \, dx \right]$$

$$= e^x \cos x + e^x \sin x - \int e^x \cos x \, dx$$

$$\therefore 2\int e^x \cos x \, dx = e^x \sin x + e^x \cos x$$

$$\int e^x \cos x \, dx = \frac{e^x}{2}(\sin x + \cos x) + c$$

$u = \cos x$	$\frac{dv}{dx} = e^x$
$\frac{du}{dx} = -\sin x$	$v = e^x$
$u = \sin x$	$\frac{dv}{dx} = e^x$
$\frac{du}{dx} = \cos x$	$v = e^x$

Note that the integral recurs on the RHS after doing integration by parts twice.

CONCLUSION

In this chapter, we learnt about the various techniques of integration. Integration is essentially the reverse process of differentiation, and this relationship provides us with a way to check our answers. The next chapter will formally introduce the concept of definite integrals and how the evaluation of definite integrals helps us with the calculation of areas and volumes of geometrical objects.

SUMMARY